TSNM

TOUCH SENSING NOTE MEMORY **v4.1**

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MMXXII

Table of contetnts 3

Live, settings, internal clock 4

Keyboard and arpeggiator settings 5
Arp state, hold 6
Arp direction 6
Arp variation 6
Arp clock 7
Arp rhythm 7
Behaviour of transpose buttons 8
Pressure transpose 8

Note sequencer settings 9
Sequencer state 10
Sequence recorder 10
Edit step 11
Sequence length 11
Sequence direction 11
Note variation 12
Sequencer clock 12
Sequencer reset 13
Gate pressure 13
CV mode 13

Pressure output settings 14
Pressure output 15
Pressure recorder 16
Attack duration, shape 17
Dec./rel. duration, shape 17
Modulation of attack duration 17
Modulation of dec./rel. duration . . . 17
Modulation of envelope level 18
Envelope gate source, hard reset . . . 18
Pressure slew limiter 18
Pressure recorder settings 19
Sequence length 19
Sequencer clock 19

Quantizer settings 20
Scale 20
Root note 20

Portamento settings 21
Portamento time 21
Portamento shape 21
Portamento mode 21

Vibrato settings 22
Vibrato amount 22
Vibrato speed, shape 22

Gate and trigger output settings 23
Gate output source 24
Trigger output source 24

Presets 25

Calibration 26
Note output calibration 27
Pressure output calibration 27
Volt/octave 28
Init DAC values 28
Pressure bottom 28
Pressure top 29
Save calibration 29
Exit calibration 29

Live

Usually this is the place where you want to perform. The new firmware adds some more possibilities. Press and hold the encoder knob and immediatly rotate it to toggle between functions of live page. This may feel quirky for the first time, but makes easier to reach some important paramteres of the module (listed below). Press and hold encoder knob for 1 second to enter settings.

Plate note values: Press and hold a note plate and rotate the encoder to select a note (the Volt/octave output is quantised to semitones or to a selected scale). The same action also applies to transposition buttons. Their value can range from -36 to +36 semitones. Hold both transposition buttons to edit the transposition distance when none of them is activated.
Arp state: Rotate the encoder to disable/enable arpeggiator.
Arp hold: Rotate the encoder to disable/enable hold.
Note sequencer state: Rotate the encoder to disable/enable note sequencer.
Note sequencer recorder: Rotate the encoder to enable/disable recording, when enabled the recording mode will automatically switch to overdubbing.
Pressure recorder: Rotate the encoder to enable/disable recording, its behavior will correspont to the set parameters.

Settings

Rotate the encoder to scroll through main groups of parameters, click the knob to enter a group. When entered rotate the encoder to scroll through parameters, hold one (or both) of transpose buttons to edit parameters. To exit a group of settings click the encoder knob, to leave settings press and hold encoder for 1 second.

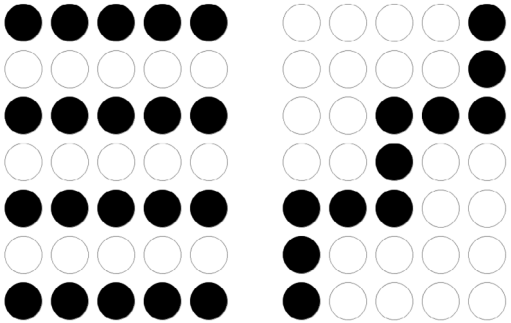
LT means: *hold the left transpose button and rotate encoder to edit parameter*

BT means: *hold both transpose buttons and rotate encoder to edit parameter*

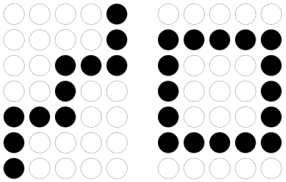
RT means: *hold the right transpose button and rotate encoder to edit parameter*

Internal clock, manual reset

The new firmware adds an internal clock, its speed can be tapped by holding the left transpose button and clicking the encoder knob. In live mode you may want to avoid this action, since it will toggle the state of transposition. To reset arpeggiator and sequencer hold the right transpose button and click the encoder.



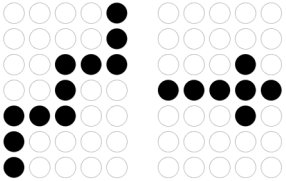
KEYBOARD AND ARPEGGIATOR SETTINGS



Arp state, hold

LT *disable/enable* arpeggiator

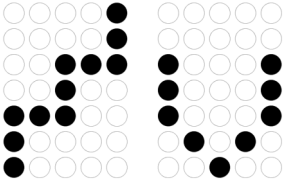
RT *disable/enable* hold, when enabled the arpeggiator will continue play the arpeggiated sequence even when note plates are not touched



Arp direction

LT direction of arpeggiation:

- *up*
- *down*
- *up-down*
- *random*



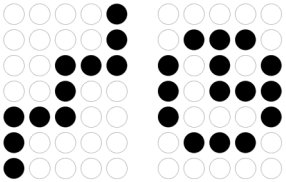
Arp variation

LT status and behavior:

- *off*
- *add variation after the arpeggiated sequence ends*
- *add variation after each arpeggiated note*

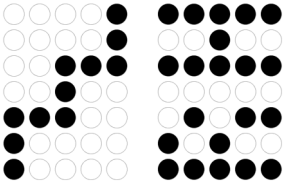
BT distance of added notes *(1 to 12 semitones)*

RT number of variations *(1 to 5 times)*



Arp clock

LT individual clock rate of arpeggiator, which can be the division or multiplication of the internal clock/external sync *(:32, :24, :16, :12, :8, :6, :4, :3, :2, *1, *2, *3, *4, *6, *8)*



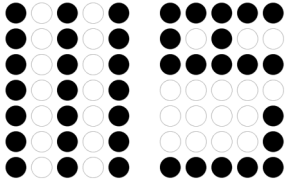
Arp rhythm

Applies a rhythmic pattern to the arpeggiators clock source.

LT length of pattern *(2-16)*

BT select step to edit

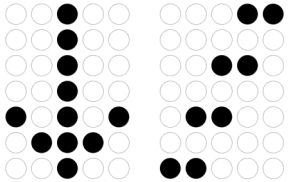
RT turn selected step *off-on*



Behaviour of transpose plates

LT behaviour:

- *latching* (takes effect as soon a note plate is touched)
- *latching immediate* (takes effect immediately)
- *momentary* (takes effect till it's touched)
- *octave* (from -3 to +3 octave)



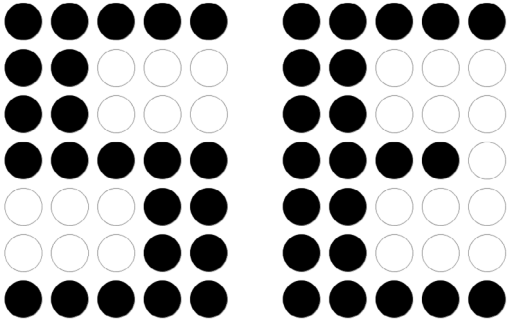
Pressure transpose

When enabled the outgoing note gets transposed as pressure is applied to note plates. This feature is most usefull when both the arpeggiator and sequencer are disabled. The basic idea comes from Skot Wiedmann's Hyve synthesizer.

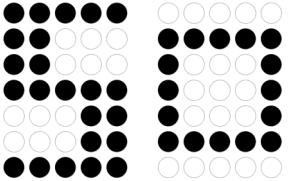
LT status and number of transpositions:

- *x- [off]*
- *x2*
- *x4*
- *x8*

RT distance of transposition *(1 to 12 semitones)*

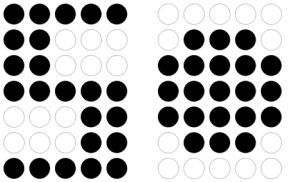


NOTE SEQUENCER SETTINGS



Sequencer state

LT *disable/enable* sequencer



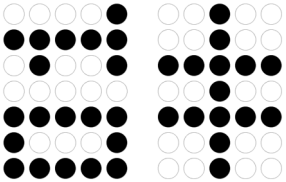
Sequence recorder

LT press to *disable/enable* recording

RT mode of recording:

– *new sequence:* when recording is enabled the notes of the sequence are entered by playing the note plates, to skip a step forward press the upper transpose plate or rotate the encoder CW, to undo a step (move back) rotate it CCW

– *overdub:* when recording is enabled the sequencer keeps playing, to overwrite the programmed melody just hit the note plates (ratcheting gate variations are pressure sensitive), press upper transpose plate to mute gates per step



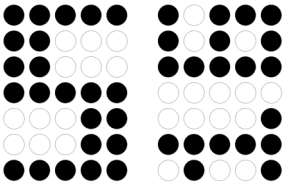
Edit gate/note per step

LT select step *(1 to sequence length)*

BT set note per step

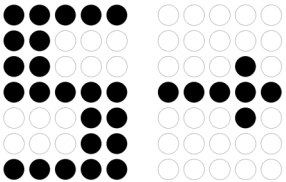
RT set gate type of selected step:

- *----------* *(off)*
- *IIIIIIIIIII (full on)*
- *IIIIII------*
- *III---III---*
- *II--II--II--*
- *I--I--I--I--*



Sequence length

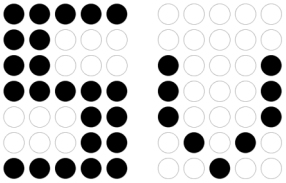
LT number of steps in the sequence *(2 to 32)*



Sequencer direction

LT direction of sequencing:

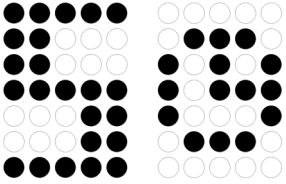
- *forward*
- *backward*
- *ping-pong*
- *random*



Note variation

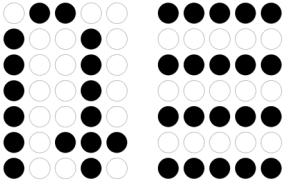
LT probability, that a random note will be added to the sequence *(0% to 99%)*

RT scope of random notes, the maximum distance of random notes compared to the pre-programmed notes of sequence *(2 to 32 semitones)*



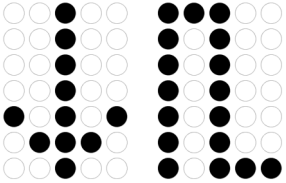
Sequencer clock

LT individual clock rate of sequencer, which can be the division or multiplication of the internal clock/external sync *(:32, :24, :16, :12, :8, :6, :4, :3, :2, *1, *2, *3, *4)*



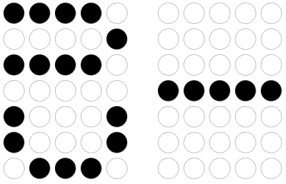
Sequencer reset

LT *disable/enable* reset from note plates (when enabled the sequencer will restart [reset its playhead] when a new note is played on the note plates)



Gate pressure

LT *disable/enable* pressure control of active gates (when enabled pressure applied to note plates will influence the pre-programmed gate types of steps)

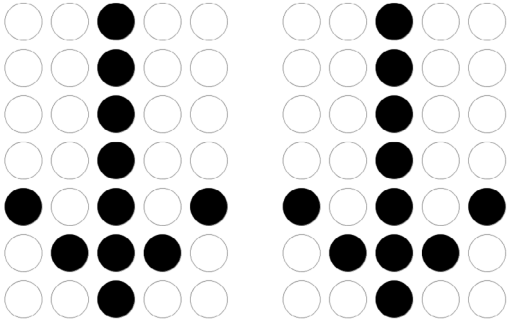


CV mode

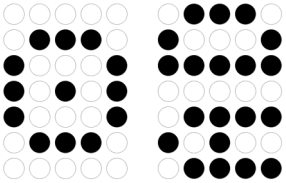
LT behavior of sequencer note CV output:

- *continuous:* note output will change according to the programmed note values per step

- *gated*: note output will be updated only when a step has non-off gate type



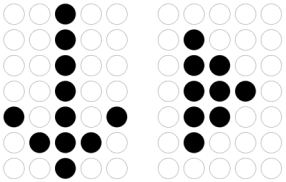
PRESSURE OUTPUT SETTINGS



Pressure output

LT available sources:

- *`pressure`*: voltage on the pressure output will continuously follow the pressure applied to note plates
- *`velocity`*: voltage on pressure output will represent the force with which a note plate is touched (it takes some time to measure the velocity, therefore it's delayed by a few milliseconds compared to the event)
- *`AD:`* attack/decay envelope generator
- *`AR:`* attack/sustain/release envelope generator
- *`LO:`* looped AD envelope generator
- *`recorder:`* tape or sequencer output
- *`random:`* outputs a random voltage level, runs from the same trigger source as the envelope generator
- *`poly1:`* two note polyphony, the pressure output acts as the note cv output of second note
- *`poly2:`* same as above, but with a different behavior of the polyphony algorithm (each new note appears on the main note output, the previous note will moved to the pressure output)



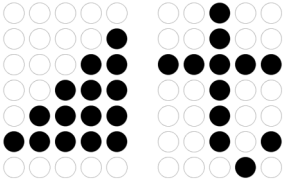
Pressure recorder

Press a note plate to record pressure information.

RT mode of recording:

- *new sequence/tape:* pressing a note plate will restart the tape/sequence, the duration of contact determines the length of tape (the length of sequence is set by *sequence lenth* parameter, see below)

- *overdub:* pressing a plate will momentarily enable recording, overwriting the previously recorded data on tape or value of current step of sequencer

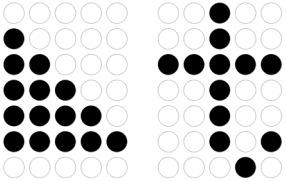


Attack duration, shape

LT coarse adjustment of duration *(0 to 63)*

BT vernier of duration *(0 to 3)*

RT shape of segment *(exp, lin, log)*

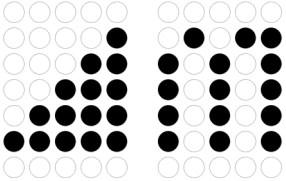


Decay/release duration, shape

LT coarse adjustment of duration *(0 to 63)*

BT vernier of duration *(0 to 3)*

RT shape of segment *(exp, lin, log)*

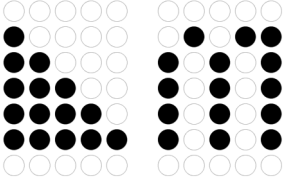


Modulation of attack duration

LT amount of modulation *(0 to 63)*

BT polarity of modulation *(+, -)*

RT source of modulation *(pressure, velocity, recorder, random)*

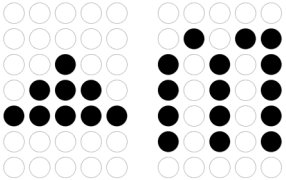


Modulation of dec./rel. duration

LT amount of modulation *(0 to 63)*

BT polarity of modulation *(+, -)*

RT source of modulation *(pressure, velocity, recorder, random)*



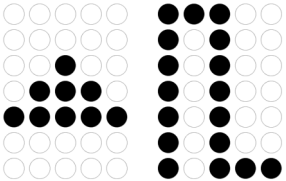
Modulation of envelope level

LT amount of modulation *(0 to 63)*

BT mode of modulation

- *T:* increasing modulation attenuates the peak (top) of envelope
- *B:* increasing modulation shifts the fundament (bottom) of envelope

RT source of modulation *(pressure, velocity, recorder, random)*

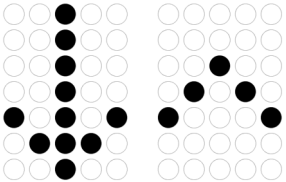


Envelope gate source, hard reset

LT source of gate sygnal that controls envelope generator:

- *automatic*
- *note plates (keyboard)*
- *arpeggiator*
- *note sequencer*

UT envelope generator hard reset *(off-on)*

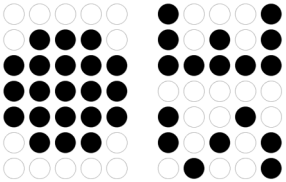


Pressure slew limiter

Creates slew limiting effect on pressure, slows down the changes of outgoing voltage.

LT rise speed *(0 to 99)*

RT fall speed *(0 to 99)*



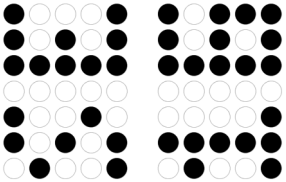
Pressure recorder settings

LT recorder mode:

- *SE (sequencer)* records pressure data in sample'n'hold-ish fashion, stores in a (maximum 32 step long) sequence
- *TA (tape)* records pressure data continuously, stores on a 5,5 second long virtual tape

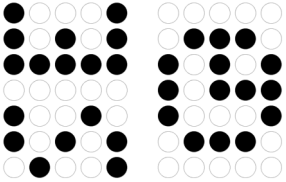
RT recorder playmode *(looping or one-shot)*

BT reset source, mandatory when playmode is set to one-shot *(no source, keyboard, arpeggiator, note sequencer)*



Sequence length

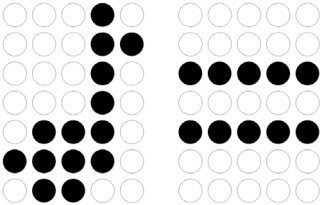
LT length of recorder sequence *(2 to 32)*

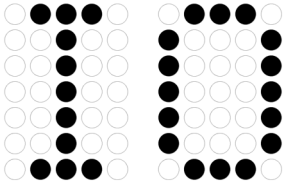


Sequence clock

LT individual clock rate of sequencer, which can be the division or multiplication of the internal clock/external sync *(:32, :24, :16, :12, :8, :6, :4, :3, :2, *1, *2, *3, *4, *6, *8)*

QUANTIZER SETTINGS



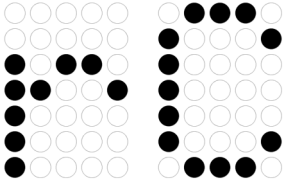


Scale

LT select scale:
Semitones, Ionian, Dorian, Phrygian, Lydian, Mixolydian, Aeolian, Locrian, Blues major, Blues minor, Pentatonic major, Pentatonic minor, Folk, Japanese, Gamelan, Gypsy, Arabian, Flamenco, Whole tone, User's Scale 1 to User's Scale 8

Editing user's scales:
BT scroll through semitones *(C to H)*

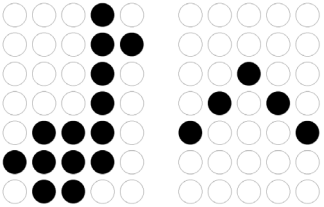
RT *disable/enable* selected semitone
User's scales are not stored with presets, they are global. To save changes go to calibration settings, scroll to *SAVE*, hold **LT** and click the encoder button.

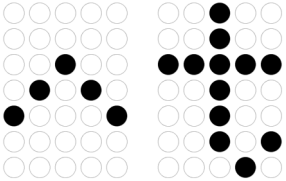


Root note

LT root note of selected scale *(C to H)*

PORTAMENTO SETTINGS

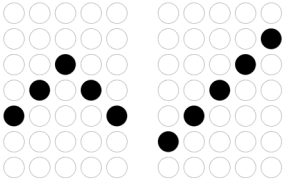




Portamento time

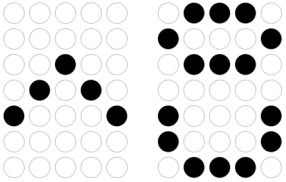
Creates slew limiting effect on the CV output, sets the speed of gliding (ie: how much time it takes to transition from one pitch to the next). The glide time will stay the same between notes, regardless of the interval.

LT time *(0 to 31)*



Portamento shape

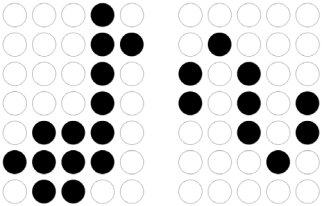
LT shape:
- *exponential*
- *linear*
- *logarithmic*

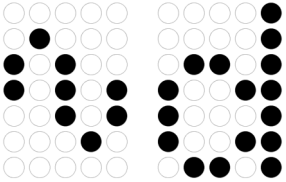


Portamento mode

LT mode:
- *continuous*
- *legato* (gliding happens only when two or more note plates are touched simultaneously)

VIBRATO SETTINGS



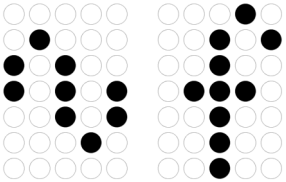


Vibrato amount

LT amount of modulation, maximum depth *(0 to 63)*

BT vernier of modulation amount *(0 to 3)*

RT pressure control over modulation depth *(off-on)*

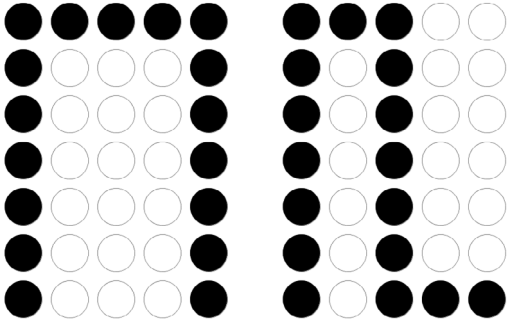


Vibrato speed, shape

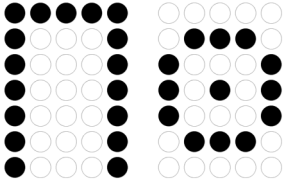
LT speed of LFO *(0 to 63)*

BT vernier of speed *(0 to 3)*

RT shape *(triangle, square, smooth random, stepped random, positive offset, negative offset)*



GATE AND TRIGGER OUTPUT SETTINGS

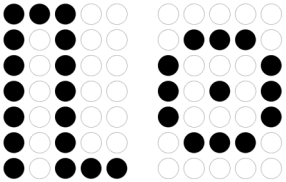


"Gate" output source

LT source:
- *automatic*
- *keyboard gate*
- *keyboard trigger* (great for exciting filters)
- *arp gate* (when arp is disabled the module will output gate from keyboard)
- *note sequencer gate* (when note sequencer is disabled the module will output gate from keyboard)
- *pressure* (turns on when pressure reaches half its range)
- *random gate* (running at 2x the speed of internal clock)
- *clock* (internal clock multiplied by 4)

RT NOT *off-on* (inverts output)

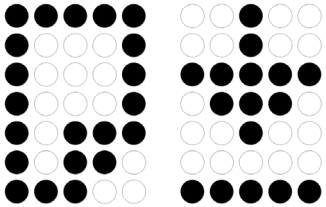
BT AND *off-on* (output will present only when note plates are touched)

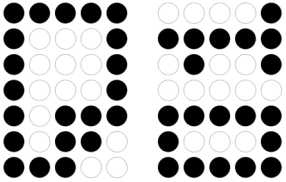


"Trigger" output source

Same as above, except:
- when *arp gate* is selected and arp is disabled the output will be muted
- when *note sequencer gat*e is selected and the sequencer is disabled the output will be muted
- clock *outputs* internal clock multiplied by 2

PRESETS





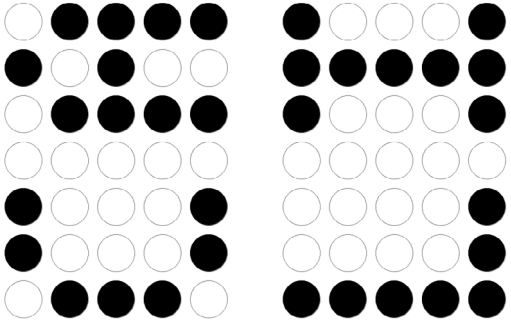
Presets

Rotate encoder to scroll through presets *(1 to 17)*.

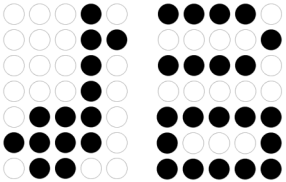
LT press encoder knob to *LOAD* selected preset

BT press encoder knob to *INIT* current parameters

RT press encoder knob to *SAVE* selected preset



CALIBRATION

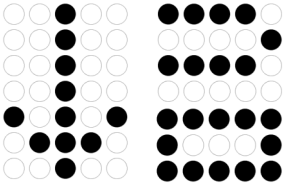


Note output calibration

Connect a multimeter to **Note CV output**, scroll through the values *(0V, 2V, 5V, 8V)* by rotating the encoder.

LT fine adjustment

RT coarse adjustment

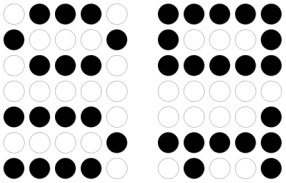


Pressure output calibration

Connect a multimeter to **Pressure CV output**, scroll through the values *(0V, 2V, 5V, 8V)* by rotating the encoder.

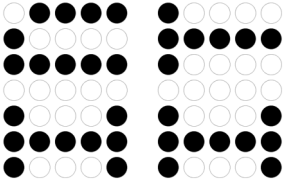
LT fine adjustment

RT coarse adjustment



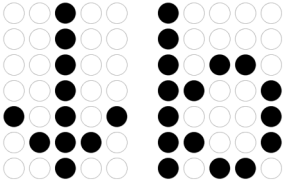
Volt/octave

LT output mode:
- *1V/oct*
- *1.2V/oct*
- *2V/oct*



Init DAC values

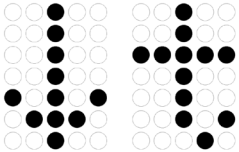
LT press encoder knob to revert DAC values to an initial state. This may be usefull, if for some reasons the program won't load them automatically at the first startup in case of fresh builds.



Pressure bottom

Apply a small amount of pressure to any note plate and press the encoder knob. The module will capture the amount of pressure and treat it as the minimal pressure value.

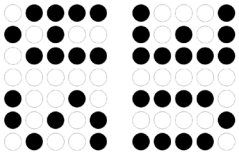
LT manual adjustment



Pressure top

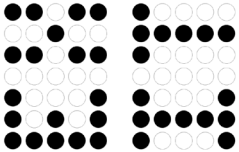
Apply the as much amount of pressure as it feels natural to any note plate and press the encoder knob. The module will capture the amount of pressure and treat it as the maximum pressure value.

LT manual adjustment



Save

Press and hold any transpose button then press the encoder to save settings.



Exit

Press the encoder to exit without saving settings.

SENSITIVITY, DEBOUNCE & ENCODER DIRECTION

To adjust these attributes press and hold the encoder during power-on, rotate it to scroll through parameters. Click the encoder to enable/disable editing of values. After setting these values the module needs to be rebooted by turning off then on your system.

Some MPR121 breakout boards may behave differently, regarding the sensitivity of touch plates. If your module in keyboard mode (with diasbled arpeggiator and sequencer) behaves like a totally unpredictable sequencer clocked by a Geiger counter try to lower **SENSITIVITY** and/or increase **DEBOUNCE**. Try to keep **SENSITIVITY** as high as possible and **DEBOUNCE** as low as possible.

The third parameter determines the **direction of encoder rotation**.

The fourth parameters enables/disables the **internal voltage refernce** of the DAC (digital-to-analog converter). Set **VR** to **IN** if using DAC8562 or DAC8563, set **VR** to **EX** if using AD5663.

If you have any questions or remarks
please contact me via e-mail:

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